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The Editors
Construction and Building Materials
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Dear Editors,

I submit the manuscript “A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments” for consideration as a research article in Construction and Building Materials.

Oyster-shell powder is promoted as a low-carbon cement substitute, but the claimed joint carbon and durability benefit is usually asserted rather than tested. The manuscript asks, under stated uncertainty, when shell-derived concrete delivers a simultaneous carbon and service-life advantage over ordinary Portland cement concrete and where that claim fails. It separates uncalcined shell powder replacing cement (P1) from shell calcium carbonate replacing limestone in the kiln feed (P2) and couples hydration, porosity, chloride, acid and leaching transport, damage accumulation and cradle-to-gate carbon in paired Monte Carlo analyses of a pavement, a dam and a marine culvert, with Sobol sensitivity and an explicit falsification region.

The paper tests the model out of sample. The inert-filler strength model is rejected by published scallop-shell concrete data (Abbas and Kudukkan 2025): it predicts relative strengths of 0.79, 0.61 and 0.43 for three mixes against 1.00, 0.86 and 0.75 observed. A filler-efficiency parameter was therefore calibrated by rejection approximate Bayesian computation on two mixes (posterior median 0.65) and tested on the third, and the main claims were re-evaluated under the calibrated posterior. Calibration changes the conclusions: for the pavement at 25% replacement the probability of a carbon cut of at least 5% with non-inferior life rises from 0.01 to 0.23, while it stays at 0.00 in the marine culvert. The results are therefore reported as conditional on the strength calibration and on a chloride-transport model that remains uncalibrated, and the experiments that would test them are specified.

A companion manuscript prepared for the Journal of Cleaner Production treats the competing use of recovered shell, reef restoration, and allocates shell between reef, concrete and disposal; this paper is restricted to the concrete pathways.

Code, automated checks, all numerical results and the Supplementary Information are openly available at https://github.com/sandlerleon/oyster-shell-concrete-mc and archived at Zenodo (software, doi:10.5281/zenodo.23867048; manuscript preprint, doi:10.5281/zenodo.23867044). The original model was re-run from a clean directory before any change and its results reproduced to numerical precision.

No new experiments are reported; the literature screen supporting the stated gap is limited and is described as such.

A generative AI assistant (Claude, Anthropic) was used to help develop and test the simulation code, to run the analyses and to draft and revise the manuscript text and figures, as declared in the manuscript. I specified the research questions, reviewed and edited all content and take full responsibility for it. The manuscript is original, has not been published elsewhere and is not under consideration at any other journal. I am the sole author, I have no competing interests, and the work received no specific funding.

Thank you for considering this submission.

Sincerely,

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